



A retrospective study of 29 fatal cases of insulin overdose

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ARTICLE INFO

Keywords:

Insulin
Insulin overdose
Forensic toxicology
Case review

ABSTRACT

Purpose: To summarize recent cases of fatal insulin poisoning both domestically and internationally, thereby offering valuable insights for the forensic identification of insulin overdose cases.

Methods: Literature published since 2000 on fatal insulin overdose were systematically searched and screened. Data encompassing variables such as year, age, sex, cause of death, scene conditions, occupations, medical histories of victims and perpetrators, autopsy timing, dosage and administration methods, forensic pathology, and toxicological analysis, were compiled for rigorous statistical analysis.

Results: Among the 29 fatal cases of insulin poisoning, suicides and homicides accounted for 55.2 % and 41.4 %, respectively. Precisely 34.5 % of victims or perpetrators were associated with the medical industry, 27.6 % had diabetes, and 24.1 % had mental illnesses such as depression. Intravenous injection resulted in quicker death than did subcutaneous injection. In some cases, immunohistochemical staining of insulin and protamine at injection sites yielded positive results. The average molar ratio of insulin to C-peptide in post-mortem blood was 13.76 ± 5.167 , indicating a significant diagnostic value for insulin poisoning.

Conclusion: Assessment of cases of fatal insulin overdose should be thorough, incorporating case investigation, scene examination, medical records review, autopsy findings, pathological examinations, and laboratory tests, alongside considering the condition of the body and timing of death autopsy. Using mass spectrometry to detect insulin proves valuable, particularly in cases of poor body preservation.

1. Introduction

Insulin is an important protein hormone with various synthetic and metabolic physiological functions. Human insulin and synthetic insulin analogs are widely used in the treatment of type 1 and type 2 diabetes. However, excessive use of insulin can lead to severe hypoglycemia, followed by irreversible neuronal necrosis, ultimately resulting in death [1,2]. Due to its peculiar nature of administration, insulin is often utilized for suicide and criminal activities [3,4]. Post-mortem identification of insulin overdose can be challenging due to factors such as hemolysis and degradation that rapidly decrease insulin levels in post-mortem blood [5]. This study is based on the identification cases completed by our unit, as well as a review of insulin overdose literature from domestic and international sources since 2000. The collected information includes the year, age, sex, cause of death, scene findings, victim and perpetrator professions and medical histories, post-mortem interval, injection dosage and method, histopathology, autopsy, and toxicological analysis results. We aimed to provide references for forensic identification of

insulin overdose cases.

2. Methods

Information on insulin overdose cases was obtained from literature searches and identification cases completed by our unit. Chinese cases were searched on the China National Knowledge Infrastructure (CNKI) and WanFang Data using keywords such as insulin overdose, massive insulin, insulin suicide, and insulin homicide. Foreign cases were searched on PubMed using keywords such as insulin toxicity, insulin overdose, insulin poisoning, insulin suicide, and insulin homicide. The inclusion criteria were original studies on fatal cases of insulin overdose since the year 2000 and cases confirmed by our unit as deaths due to insulin overdose. The exclusion criteria were cases lacking crucial background information. A total of 29 fatal cases of insulin overdose were obtained (Table 1).

Data are presented as mean $\pm$ standard error of the mean (SEM). All data were analyzed using GraphPad Prism 8.0 software. Differences

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Table 1
Characteristics of the 29 fatal cases of insulin overdose.

Case No.	Year	Age (years)	Sex	Manner of death	Victim and murderer information	Items found at the scene	Injection method	Survival time after injection	Total injection dosage	Injection site
1[6]	2000	68	M	Suicide	Suicidal doctor; depression	3 insulin syringes, 1 insulin bottle	Subcutaneous	/	/	Abdomen
2[7]	2000	24	M	Suicide	Student	/	Subcutaneous	< 10 h	/	Left arm
3[7]	2000	44	M	Suicide	/	Novolin® 70/30, 1 syringe	Subcutaneous	< 20 h	/	/
4[8]	2001	25	M	Homicide	Killed lover; the murderer was a nurse	/	Intravenous	1 h	1200 U	Left arm
5[9]	2002	5	F	Homicide	Father killed young children; the murderer was a pharmacist	Insulin aspart 40 R	Subcutaneous	6 h	70 U	Right forearm
6	2007	22	F	Homicide	/	/	Intravenous	< 14 h	/	/
7	2010	27	F	Homicide	Husband killed wife; the murderer was a doctor	/	Intravenous	1 h	/	/
8	2011	45	M	Homicide	Wife killed husband; the murderer had diabetes	Insulin bottle	/	40 h	>1500 U	/
9	2012	28	M	Homicide	Killed lover; the murderer was a nurse	/	Subcutaneous	< 2 h	600 U	Left hand and left arm
10[10]	2012	55	F	Homicide	/	Insulin syringes and bottles	/	4 days in hospital	/	/
11[11]	2013	62	F	Suicide	The suicide victim had thyroid cancer; her husband had diabetes	Insulin bottles	Subcutaneous	7 h	< 800 U	/
12[12]	2015	27	M	Homicide	Killed boyfriend; the murderer was a nurse	Insulin bottles	Subcutaneous	10 h	400 U	Left elbow and right abdomen
13[12]	2015	55	M	Homicide	Wife killed husband; the murderer had diabetes	10 insulin pens	Subcutaneous	< 8 h	1500 U	Left arm and abdomen
14[13]	2015	68	F	Suicide	Suicidal doctor; depression	/	Subcutaneous	/	/	Right thigh
15[12]	2016	85	F	Homicide	Daughter killed mother; the victim had diabetes	/	Subcutaneous	< 6 h	1200 U	Abdomen
16[14]	2016	29	F	Suicide	Suicide, had diabetes and depression	Insulin glargine and aspart	Subcutaneous	< 8 h	< 560 U	/
17	2018	36	F	Suicide	/	Insulin pens, bottles, and a suicide note	Subcutaneous	5 days in hospital	1500 U	Left wrist, left forearm, abdomen
18	2019	32	F	Suicide	Endocrinologist; was depressed	5 insulin syringes and a suicide note	Subcutaneous	15 h	> 1500 U	Abdomen
19[15]	2019	38	F	Suicide	/	Bottles of insulin lispro,	Subcutaneous	< 8 h	/	Left abdomen
20[16]	2019	68	M	Suicide	The suicide victim was a hospital emergency worker with depression	2 bottles of insulin human	Sublingual	/	< 600 U	/
21	2020	22	M	Homicide	Mother killed son; the victim was schizophrenic	/	Subcutaneous	34 h	900 U	Left elbow, abdomen
22	2020	39	F	Homicide	Killed lover, the murderer is a retired soldier	/	Subcutaneous	6 h	300 U	Abdomen
23[17]	2021	47	F	Suicide	/	Insulin	/	/	/	/
24[17]	2021	36	M	Suicide	/	Insulin	/	/	/	/
25	2022	45	F	Accident	The victim had diabetes	/	Intravenous and subcutaneous	< 2 h	14 U (Postoperative fasting period)	Right cubital fossa, right hand
26[18]	2022	30	M	Suicide	Suicide; had diabetes	8 insulin bottles	Subcutaneous	9 h	< 2400 U	Abdomen
27[19]	2023	64	M	Suicide	Suicide; had diabetes	Four pens of insulin injection devices (NovoRapid) and a suicide note	Subcutaneous	/	< 1200 U	Left elbow and left wrist
28[20]	2023	26	F	Suicide	The suicide victim was a nurse	Insulin syringes and a suicide note	Intravenous	/	/	Vein catheter in left wrist
29[21]	2023	30	F	Suicide	Suicide; had diabetes and psychosis	3 bottles of insulin glargine and 6 bottles of insulin lispro	Subcutaneous	7 h	< 2700 U	Abdomen
No.	Post-mortem interval		Detection method		Insulin concentration	C-peptide concentration	Insulin/C-peptide	Glucose concentration	Histopathology	
1	1 day		RIA		PMB 1848.8 µU/mL VH 6.9 µU/mL	< LOD	PM > 44	/	/	

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Table 1 (continued)

No.	Post-mortem interval	Detection method	Insulin concentration	C-peptide concentration	Insulin/C-peptide	Glucose concentration	Histopathology
2	/	RIA	VH 31 µU/mL	/	/	VH 26 mg/dL	/
3	1 d	RIA	PMB 840 µU/mL	PM 0.5 ng/mL	PM 35	AMB 26 mg/dL	/
4	30 h	RIA	PMB 54 µU/mL	PM 166 pmol/L	PM 2.34	PMB 25.4 mmol/L	/
5	12 h	RIA	AMB 694 µU/mL PMB 295 µU/mL	AM 0.6 ng/mL PM 0.5 ng/mL	AM 3.47 PM 1.77	AMB 18 mg/dL PMB 256 mg/dL	Congestion of lungs and kidneys
6	1 day	RIA	PMB >100 µU/mL	PM 0.55 ng/mL	PM 3.8	PMB 1.2 mmol/L	/
7	2 days	/	/	/	/	/	/
8	/	/	/	/	/	/	/
9	18 days (3 days in water and frozen for 15 days)	RIA	PMB 2.14 µU/mL	PM < 0.01 ng/mL	PM 4.46	PMB 12.16 mmol/L	Liver and lung congestion, insulin-positive and protamine-positive at the injection site
10	Immediately	RIA (PM) MS (AM)	AMB 5551 µU/mL Insulin human VH 28.8µU/mL	AM 1.7–4.8 ng/mL	AM 3.47–9.81	AMB 2 mg/dL	/
11	Immediately	MS	Insulin lispro VH 103µU/mL Injection site 581, 808 µU/g Muscle 373 µU/g Kidney 384 µU/g	/	/	/	/
12	1 day (room temperature)	RIA	PMB 0.2 µU/mL Injection site 426.3 µU/g	PM < 0.01 ng/mL	PM 0.42	PMB 9.1 mmol/L	Insulin positive at the injection site
13	4 days	RIA	Injection site 67.2 µU/g	/	/	/	Pulmonary hemorrhage, pulmonary congestion, hepatic congestion, reactive proliferation of pontine astrocytes
14	18 h	RIA	PMB 582.5 µU/mL VH 11.5 µU/mL Bile 19.8 µU/mL PF 67.8 µU/mL CSF 17.3 µU/mL	PM 0.1 ng/mL	PM 17.5	/	Insulin positive at the injection site
15	10 days	RIA	PMB 2.5 µU/mL Injection site 121100 µU/g	PM 0.1 ng/mL	PM 0.52	PMB 2.7 mmol/L	Pulmonary congestion
16	Immediately	MS	Insulin aspart PMB 4741 µU/mL	/	/	/	/
17	1 day	RIA	AMB 4.9 µU/mL Injection site 15.9–131.99 µU/g CSF 2.3 µU/mL PF 11.6 µU/mL Bile 12.0 µU/mL VH 6.4 µU/mL	AM < 0.01 ng/mL	AM 10.22	AMB 0.9 mmol/L	Brain edema, extensive neuronal necrosis, significant damage to hippocampal neurons, pulmonary edema, insulin-positive at the injection site
18	/	/	/	/	/	/	/
19	Immediately	MS	Insulin lispro VH 2 ng/mL	/	/	/	/
20	2 days	RIA	Right VH 61.11 µU/mL Left VH 74.23 µU/mL	Right VH 0.166 ng/mL Left VH 0.157 ng/mL	/	VH < 0.1 mg/dL	Pulmonary hemorrhage, edema, proliferation of subcortical and corpus callosum astrocytes
21	1 day	RIA	PMB 15.6 µU/mL Urine 118 µU/mL Injection site 388.5 µU/g	PM 0.08 ng/mL Urine 0.04 ng/mL	PM 4.07	/	Pulmonary congestion and edema, insulin-positive and protamine-positive at the injection site

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Table 1 (continued)

No.	Post-mortem interval	Detection method	Insulin concentration	C-peptide concentration	Insulin/C-peptide	Glucose concentration	Histopathology
22	1 day (room temperature)	RIA	VH 6.3 μ U/mL Urine 33.9 μ U/mL Injection site >3000 μ U/g	VH < 0.01 ng/mL	/	/	Brain edema, neuronal necrosis, reactive proliferation of astrocytes, pulmonary congestion and edema, insulin-positive and protamine-positive at the injection site
23	< 2 days	MS	Insulin aspart FB 96 ng/mL	/	/	/	/
24	< 2 days	MS	Insulin aspart FB 432 ng/mL	/	/	/	/
25	1 day	/	/	/	/	/	Cerebral congestion, edema, pulmonary congestion
26	1 day	RIA	Injection site 200000 μ U/g Left CB 163 μ U/mL Right CB 374 μ U/mL FB 5640 μ U/mL VH 44.7 μ U/mL CSF 41.2 μ U/mL Bile 0.57 μ U/mL PF 37.9 μ U/mL Pancreas 665 μ U/g	Left CB 0.06 ng/mL Right CB 0.12 ng/mL	Left CB 8.16 Right CB 9.36	Left CB 2 mg/dL Right CB 1 mg/dL	Pulmonary congestion, edema, insulin-positive at the injection site
27	/	MS	Insulin aspart PMB 58–71 ng/mL	/	/	/	/
28	/	MS	Insulin glargine PMB 1.06 ng/mL	/	/	/	/
29	3 days	MS	Insulin glargine FB 429 μ U/mL CB 1362 μ U/mL	/	/	PMB 6 mg/dL	Insulin-positive at the injection site

The source of each case literature is cited after the No., and some cases from China have not been published in international journals. Information not reported in the original text is represented by the symbol /. Due to the influence of testing methods, the insulin concentration units reported in various literature are inconsistent, and the units in the original literature data are retained in this table. To calculate the molar ratio of insulin/C-peptide, we first converted the units, where 1 ng/mL c-peptide = 333 pmol/L; 1 μ U/mL insulin = 6.945 pmol/L. AMB: antemortem blood, PMB: post-mortem blood, VH: Vitreous humor, CSF: cerebrospinal fluid, CB: cardiac blood, FB: femoral blood, PF: pericardial fluid, RIA: radioimmunoassay, MS: Mass spectrometry.

between groups were compared using independent sample t-tests.

3. Results

3.1. Background information

Fig. 1 shows the age distribution of cases of insulin overdose. Among the suicide victims, no individual was aged 0–18, nine were aged 18–40, two were aged 40–60, and five were > 60 years. Among the homicide victims, one was aged 0–18, seven were aged 18–40, three were aged 40–60, and one was > 60 years. No significant differences were observed in terms of age and sex distribution among the study population ($p=0.903$). Age ($p=0.293$) and sex ($p=0.295$) had no significant impact on the mode of death.

3.2. Manner of death

Fig. 2 shows the mode of death in cases of insulin overdose. Among all 29 cases, 16 were suicides (55.2 %), 12 were homicides (41.4 %), and one was a medical accident (3.4 %).

3.3. Victim and murderer information

Among 29 cases, nine (27.6 %) victims or murderers had diabetes, including five suicides, three homicides, and one medical accident. Seven (24.1 %) victims had depression or mental illness, including six suicides and one homicide. Ten cases (34.5 %) of victims or perpetrators were related to the medical industry, including five cases each of suicide and homicide. All homicide cases were committed by acquaintances. We were unable to obtain relevant information for some cases.

3.4. Scene

Among the 28 cases of suicide and homicide, evidence of insulin injection at the scene, including bottles, syringes, needles, purchase records, and suicide letters, was found in 19 cases (67.9 %).

3.5. Autopsy

Among the 29 cases, injection sites or residual venous pathways were observed in 20 cases (71.4 %) during autopsy. Subcutaneous injection was the most common method of insulin administration, with occasional intravenous injection and sublingual administration. The common injection sites for subcutaneous injection were the abdomen and outer thigh, followed by the gluteus maximus and deltoid muscle of the upper arm. Intravenous injection was commonly seen in the veins of the elbow and back of the hand and was also performed through indwelling venous channels. Fig. 3 shows the effect of injection methods on survival time. When the lethal dose was reached, intravenous injection resulted in

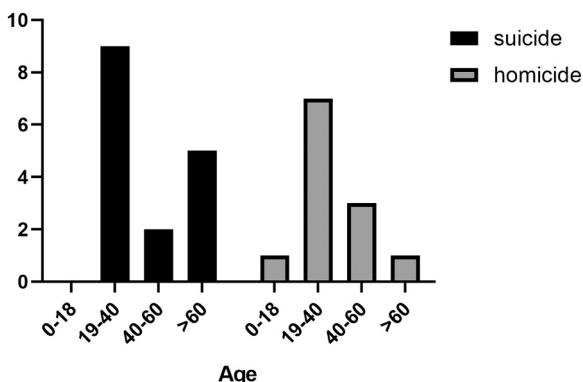


Fig. 1. Age distribution of patients with insulin overdose.

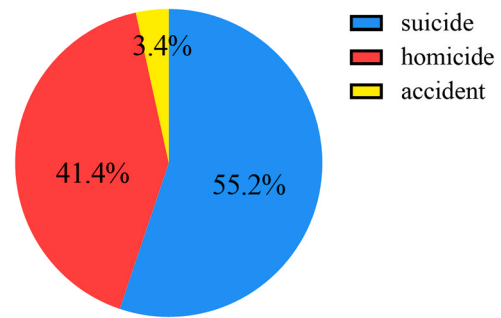


Fig. 2. Manner of death in cases of insulin overdose.

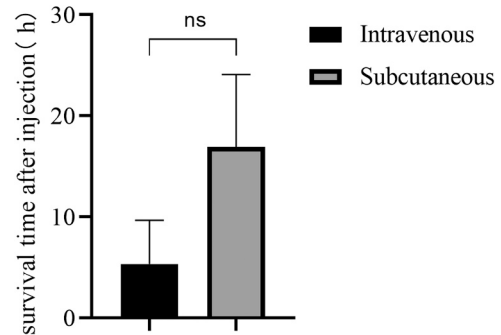


Fig. 3. Injection method and survival time.

faster death than subcutaneous injection (subcutaneous venous group $\pm$ SEM: 11.60 ± 16.95 h); however, this difference was not significant ($p=0.503$).

Among the 29 cases, 10 cases (34.5 %) had pathological changes such as pulmonary congestion and pulmonary hemorrhage. Four cases (13.8 %) showed pathological changes in the central nervous system, including neuronal necrosis and proliferation of numerous reactive astrocytes. Eight cases (27.6 %) showed positive immunohistochemical staining for insulin or protamine at the injection site. Fourteen cases (48.3 %) did not undergo pathological examination or reported detailed autopsy results, resulting in missing relevant data.

3.6. Toxicological analysis

Among the 29 cases, insulin concentrations in various biological samples were reported for 25 cases, with 17 using radioimmunoassay (RIA) and nine using mass spectrometry (MS)-related technologies. Both methods were used simultaneously in case 10. Table 2 shows the insulin content of various biological samples measured by immunological

Table 2

Insulin concentration in various biological samples in cases of insulin overdose leading to death.

Insulin concentration	Antemortem	Post-mortem
Blood	694 μ U/mL–5551 μ U/mL	54 μ U/mL–5640 μ U/mL
Injection site	/	15.9 μ U/g–200000 μ U/g
Urine	/	33.9 μ U/mL–118 μ U/mL
Vitreous humor	/	6.3 μ U/mL–74.23 μ U/mL
Bile	/	0.57 μ U/mL–19.8 μ U/mL
Cerebrospinal fluid	/	2.3 μ U/mL–41.2 μ U/mL
Pericardial fluid	/	11.6 μ U/mL–67.8 μ U/mL

Case 9 had an 18-day interval before an autopsy, case 12 was kept 1 day at room temperature before the autopsy, and case 15 had a 10-day interval before the autopsy. Meanwhile, the insulin concentration in case 17 was measured after several days of treatment. The insulin in these case samples was completely degraded and not included in the table.

methods after death due to insulin overdose.

Among the 17 cases where RIA was used, the molar ratio of insulin to C-peptide after death was calculated in 12 cases, with a result of 10.77 ± 4.133 . After excluding samples with poor preservation conditions, such as cases 9, 12, and 15, the molar ratio of insulin to C-peptide after death in nine cases of insulin overdose was 13.76 ± 5.167 .

Among the 29 cases, the blood glucose concentrations before or after death were reported in 11 cases (Fig. 4). The blood glucose concentration before death was 0.86 ± 0.28 mmol/L, ranging from 0.11–1.44 mmol/L. The post-mortem blood glucose concentration was 8.15 ± 3.15 mmol/L, with a range of 0.08–25.4 mmol/L.

4. Discussion

Several systematic studies on insulin overdose leading to death have been conducted in recent years. Marks [21] reported 66 cases of insulin murder; Bottinelli et al. [23] summarized the insulin concentration results of 31 fatal cases of insulin overdose; and Stephenson et al. [24] reported 40 cases of insulin overdose deaths from South Australia. Original information, such as case details, were not included in most of these studies; thus, were not included in this article.

Among the previous reports of insulin overdose deaths, suicide accounts for 65–90 % of cases, which is prevalent among patients with diabetes [24,25]. Among the cases we included, 55.2 % were suicides, and most of the victim and their families had medical conditions, including diabetes and depression. In cases of insulin suicide, family members often do not seek forensic identification, resulting in fewer reported cases and a serious underestimation of the number of such incidents. Insulin is also commonly used for homicide purposes, and criminals are often engaged in medical-related professions where they know and can easily obtain insulin [26]. Among the cases included in this article, homicide accounted for 41.4 % of cases, with approximately half of the offenders engaged in medical-related professions. All homicide cases are committed by acquaintances, as they can conveniently inject insulin into the victim.

The routine treatment dosage of insulin for patients with diabetes is approximately 0.2–0.5 IU/kg. There is no clear report on the lethal dose of human insulin [22]. Usually, a single injection of 2 IU/kg can lead to death. The dosage of insulin injection needs to be determined comprehensively based on the scene, medication purchase records, and the murderer's confession. Notably, the lethal dose of insulin can be affected by whether the victim has consumed food or not. In case 17, the deceased injected a large amount of insulin into her partner before committing suicide using insulin, intending to die together. However, her partner regained consciousness midway, ate half a plate of shredded potatoes, and was spared.

At the scene of suicide with excessive insulin use, bottles, syringes, needles, purchase records, and suicide letters of insulin preparations can often be found. In cases of insulin homicide, traces of insulin use may be

hidden, and attention should be paid to identifying them. Notably, excessive insulin may be combined with other causes of death. In case 4, after the victim became unconscious, an intravenous injection of 100 mL of air resulted in air embolism; in cases 7 and 13, the victims were covered with pressure on their mouth and nose after becoming unconscious, and attention should be paid when identifying insulin overdose.

Subcutaneous injection is the most common method of insulin administration [27], with occasional intravenous injection and sublingual administration. Careful autopsy usually reveals injection needle holes or indwelling venous pathways. In case 20, the deceased committed suicide using insulin preparations administered sublingually, and injection marks could not be found on the skin. Administration in this route should be performed cautiously. This study shows that intravenous injection causes death faster than subcutaneous injection. Although the difference was not significant and the effects of insulin type and injection dose were not considered, this study provides evidence that intravenous injection is more dangerous than subcutaneous injection when insulin reaches a lethal dose.

Immunohistochemical staining was performed on the injection site, and insulin positivity was mainly observed in the gaps between adipocytes, inflammatory cells, peripheral nerves, and connective tissue [12]. Additionally, protamine is more stable than insulin. If insulin preparations contain protamine, its component can often be detected at the injection site [12]. Pulmonary congestion, pulmonary hemorrhage, central nervous system neuronal necrosis, and astrocyte proliferation can also serve as references for identifying cases of insulin overdose.

RIA is currently a clinical method for detecting insulin and C-peptide. This method is more reliable for fresh or well-preserved autopsy samples. For autopsy samples that are rotten, stored for extended duration, and have undergone severe hemolysis, the results obtained by RIA are inaccurate due to the rapid degradation of insulin content in blood samples after death. Insulin levels in other biological samples, such as those collected from injection sites, urine, vitreous fluid, cerebrospinal fluid, bile, and pericardial fluid, decrease relatively slowly. In this study, case 9 had an 18-day interval before the autopsy, case 12 was kept 1 day at room temperature before the autopsy, and case 15 had a 10-day interval prior to the autopsy. The blood insulin in these cases was almost completely degraded; however, the insulin concentration detected at the injection site remains significant.

RIA cannot determine the type of insulin in the blood; however, the use of exogenous insulin can be determined by calculating the ratio of insulin to C-peptide. After the degradation of proinsulin, insulin, and C-peptide are released into the portal vein circulation in equimolar amounts [7]. Due to the slightly longer half-life of C-peptide, the molar ratio of peripheral blood insulin/C-peptide is always < 1 in healthy individuals. When injecting exogenous insulin, the molar ratio of insulin/peptide is > 1 [28]. The insulin/peptide ratio in most cases in this study ranged from 1.77–44, which can be regarded as strong evidence of exogenous insulin administration. Due to the almost complete degradation of blood insulin in the samples, the ratio of insulin/C-peptide could not be used for case determination in cases 12 and 15.

Using MS including liquid chromatography (LC)-MS [29], LC-MS/MS [30], nanoLC-MS [31], HRMS [20], ultra-performance LC-MS [32], has been widely used internationally to determine insulin levels. Moreover, it has been employed to identify cases of insulin overdose where samples have deteriorated or been stored for extended periods. Even if the insulin content is considerably low, MS can qualitatively determine the use of exogenous insulin, which is of great value for forensic identification. We reported nine cases where MS was used to accurately characterize insulin in biological samples, including types of insulin such as human, lispro, aspart, and glargine insulin.

The measurement results of post-mortem blood glucose cannot reflect the true blood glucose levels before death [33]. Peripheral blood glucose concentration decreases by approximately 1–2 mmol/L per h after death. Meanwhile, in the heart blood sample, glucose

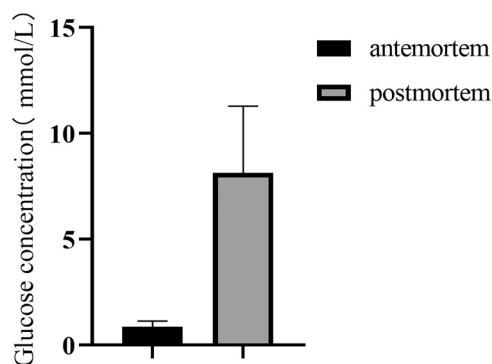


Fig. 4. Blood glucose concentration in cases of insulin overdose leading to death.

concentration may increase through enzymatic degradation of liver glycogen [3]. Additionally, due to changes in post-mortem glycolysis and other factors, the measured blood sugar may increase. Among the 29 cases in this article, four reported the results of blood glucose concentration before death, all of which were consistent with the diagnosis of hypoglycemia and had a certain reference value for the assessment of the case. However, the results of blood glucose concentration after death in eight cases were markedly broad and lacked forensic significance. As much as possible, in cases of suspected insulin overdose leading to death, clinical samples of the deceased before their death should be obtained.

5. Conclusions

Our findings revealed that insulin overdose can occur in cases of suicide, homicide, and medical accidents. Suicide cases often involve individuals with diabetes or depression. Perpetrators of homicide cases are often engaged in the medical profession and belong to groups that have knowledge of and access to insulin. Scene inspections often provide evidence of insulin injection, and autopsy usually reveals the injection site. When insulin reaches a lethal dose, intravenous injection leads to faster death than subcutaneous injection. The immunohistochemical staining results of insulin and protamine at the injection site are of great value in determining the case. Using RIA to determine insulin and C-peptide in post-mortem biological samples requires careful interpretation of the obtained results based on the preservation of the corpse; if the preservation of the body is poor, MS can be used to qualitatively test for insulin to clarify the case.

Compliance with ethical standards

The case was approved by the ethics committee of Huazhong University of Science and Technology.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Yuhao Yuan: Writing – original draft. **Yiwu Zhou:** Writing – review & editing, Writing – original draft, Conceptualization. **Yiling Li:** Data curation. **Qing Shi:** Writing – review & editing. **Shuquan Zhao:** Writing – review & editing. **Zhonghao Yu:** Writing – original draft. **Fang Tong:** Data curation.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Acknowledgments

We would like to thank Editage for assistance with English language editing.

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